

A ground-up guide to the Macro-ECL engine
Expected credit loss under IFRS 9, built on public SBA and FRED data

Project notebook / engineering notes

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Abstract

This document explains a working expected-credit-loss (ECL) pipeline from first principles. It is written so a reader who has never seen a bank risk model can follow the story, and so a quantitative reviewer can audit every formula, mapping, and limitation.

The software scores a historical book of U.S. Small Business Administration (SBA) 7(a) guarantees (Kaggle `SBAnational.csv`), attaches quarterly U.S. GDP and unemployment from FRED, fits a twelve-month probability of default (PD) and a loss-given-default (LGD) model, applies a simple IFRS 9 staging rule, and discounts cash shortfalls under three macro scenarios produced by a small vector autoregression (VAR).

The project is a **demonstration**, not a bank production system. The public tape has no GAAP financial statements, no live performing book, and no stored origination PD. Those gaps are left as limitations. They are not patched with synthetic labels or an unfitted “stress overlay.”

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Chapter 1

Who this is for, and how to read it

1.1 Two audiences, one document

If you are not a modeller. Read Chapters 2–3 and the plain-language boxes in later chapters. You should come away able to answer: *What is a provision? Why do we need scenarios? Why does this project not look like a live bank book?*

If you are a modeller or validator. Read the mapping tables, the PD/LGD specifications, the ECL recursion, Chapter 10, and Chapter 13. Every number quoted from a run is from the Kaggle+FRED path described here, not from a toy sample.

1.2 What this project is

A Python repository that:

- cleans a public SBA loan tape and a FRED macro extract;
- trains a 12-month PD classifier and a two-part LGD model;
- builds three point scenarios (baseline / adverse / severe);
- computes IFRS 9-style Stage 1 / Stage 2 / Stage 3 ECL;
- writes an SR 11-7-flavoured PDF and logs models in MLflow.

1.3 What this project is not

- Not an approved bank model.
- Not a through-the-cycle rating system.
- Not a fitted point-in-time (PIT) PD that re-prices with unemployment.
- Not a survival / competing-risk lifetime engine.
- Not a substitute for management judgement or audit.

Optional large-language-model (LLM) text can be appended to the report. It does not compute ECL.

Chapter 2

Why banks hold a provision

2.1 A loan in one paragraph

A bank lends money. Most borrowers repay. Some do not. When a borrower defaults, the bank usually recovers only part of the amount outstanding—collateral, guarantees, collections. The shortfall is a *credit loss*.

Accounting standards no longer allow the bank to wait until default is almost certain before recognising that loss. Under IFRS 9 the bank must hold an *allowance* (a provision) equal to the *expected* credit loss: a probability-weighted, discounted estimate of cash shortfalls.

2.2 The three-letter identity

Almost every credit-risk textbook starts from

$$EL = PD \times LGD \times EAD. \quad (2.1)$$

PD Probability of default over a defined horizon (here, twelve months, or lifetime).

LGD

Loss given default: the fraction of exposure that is not recovered, $1 - \text{recovery rate}$, after costs.

EAD

Exposure at default: how much is outstanding if default happens (drawn balance plus, in a full bank model, a credit-conversion factor on undrawn lines).

Equation (2.1) is an *expectation* for one period. IFRS 9 ECL is that idea stretched over time, discounted, and allowed to change when the economy or the borrower deteriorates.

2.3 Why “expected” is not a point guess

Default is a rare event for a good borrower and a common event in a recession for a weak one. IFRS 9 asks for an unbiased estimate that uses

1. past experience,
2. current conditions,
3. reasonable and supportable forecasts of the future.

In practice that means at least two or three macro scenarios with weights that sum to one. This engine uses three point scenarios and configurable weights (default 0.5/0.3/0.2 for baseline / adverse / severe).

Chapter 3

IFRS 9 in working language

3.1 Three stages, two measurement horizons

IFRS 9 puts every performing or underperforming facility into one of three *stages*. The stage decides the *horizon* of the expectation, not a different definition of default.

Stage	Horizon	Typical trigger
1	12-month ECL	No significant increase in credit risk (SICR) since origination
2	Lifetime ECL	SICR, but the asset is not credit-impaired
3	Lifetime ECL	Credit-impaired (default / 90 days past due / unlikelihood to pay)

Plain language. Stage 1 is “this still looks like what we underwrote.” Stage 2 is “this has got materially riskier; measure the whole remaining life.” Stage 3 is “this is already in trouble; expect to lose LGD on the exposure.”

3.2 SICR is a comparison, not a level

Significant increase in credit risk is supposed to compare *lifetime risk now* with *lifetime risk at origination*. Banks implement that with rating notches, PD multiples, days past due (DPD), forbearance flags, and watch-list codes.

This engine uses a transparent stand-in:

$$\text{SICR} = \left\{ \frac{\text{PD}_{12\text{m}}^{\text{now}}}{\text{PD}_{12\text{m}}^{\text{orig}}} \geq \rho \right\} \vee \left\{ \text{PD}_{12\text{m}}^{\text{now}} - \text{PD}_{12\text{m}}^{\text{orig}} \geq \delta \right\} \vee \{ \text{DPD} \geq 30 \}. \quad (3.1)$$

Default configuration: $\rho = 2.5$, $\delta = 0.08$. Stage 3 if $\text{PD}_{12\text{m}} \geq 0.60$ or $\text{DPD} \geq 90$.

On the SBA tape, DPD is almost always zero (the file is resolved outcomes, not a live arrears book). Origination PD is the same HGB scored on macros attached at origination date—there is no stored origination rating. Pre-2000 originations have no FRED history; those names use baseline PD as the SICR anchor so they are not pushed into Stage 2 by a missing macro series.

3.3 Lifetime is not a second model unless you build one

A full lifetime PD is a term structure: the chance of defaulting in year 1, year 2, ... conditional on survival, under the scenario path.

This engine uses the actuarial identity implied by a constant annual hazard h equal to the 12-month PD:

$$\text{PD}_{\text{life}}(T) = 1 - \prod_{t=1}^T (1 - h) = 1 - (1 - h)^T, \quad h = \text{PD}_{12\text{m}}. \quad (3.2)$$

That is honest and crude. It is not a Cox model and not a vintage survival curve.

Chapter 4

Architecture of the repository

4.1 Pipeline

1. **Data.** `src/data_prep/` reads raw SBA and FRED files, maps columns, attaches lagged macros as-of origination (and as-of a snapshot date).
2. **Scenarios.** `src/macro_engine/` resamples FRED to quarters, optionally runs LASSO lag selection, fits a two-variable VAR, writes baseline / adverse / severe rows to `data/scenarios/macro_forecasts`.
3. **Models.** `src/ecl_models/train_*.py` fits PD and LGD on a 70/30 stratified holdout, writes `models/*.pkl`, logs to MLflow.
4. **Engine.** `ECLExecutionEngine` scores each scenario, stages, discounts ECL, probability-weights the three totals.
5. **Report.** `SR117ReportGenerator` writes `reports/ecl_macro_SR117_Final.pdf` and a Mark-down twin.

One command:

```
PYTHONPATH=. python3 scripts/run_all.py
```

Add `--genai` only if `OPENAI_API_KEY` is set and you want LLM notes in the PDF.

4.2 Configuration

All knobs live in `config/config.yaml`: raw paths, feature lists, monotonic signs, scenario shocks, stage thresholds, scenario weights, MLflow experiment name. Changing a shock or a SICR cut should not require editing Python.

Chapter 5

The data

5.1 Why these two files

Banks will not publish a live C&I book. The two public files used here are:

Kaggle SBA National (SBAnational.csv)

U.S. 7(a) guarantees, roughly 1987–2014, on the order of 9×10^5 rows. Outcome is PIF (paid in full) or CHGOFF (charged off), plus approval and charge-off dollars.

FRED extract

Real GDP (GDPC1) and the civilian unemployment rate (UNRATE), quarterly from 2000 in the file used for this project.

5.2 What SBA National does *not* contain

No current assets, current liabilities, or total debt. No stored origination PD. No monthly arrears tape. No undrawn commitment. The book is *resolved* loans, not a snapshot of names that are still paying.

That is why mean predicted PD sits near 25% and the charge-off rate is about 18%: you are looking at finished stories, not a performing SME portfolio.

5.3 Column mapping (Kaggle layout)

When the loader sees GrAppv, SBA_Appv, MIS_Status, it uses:

Engine field	SBA rule
exposure_at_default	DisbursementGross if present, else GrAppv
default_flag	MIS_Status contains CHGOFF, or ChgOffPrinGr > 0
loss_severity	ChgOffPrinGr/EAD on defaults; 0 otherwise
leverage	$1 - \text{SBA_Appv}/\text{GrAppv}$ (unguaranteed share)
liquidity_ratio	$\log(1 + \text{NoEmp})/4$ (headcount proxy)
collateral	EAD if Term ≥ 240 months, else $0.5 \times \text{EAD}$
maturity_years	Term/12
origination_date / asof_date	approval / disbursement dates
days_past_due	0 (not on the file)

Dollar fields that arrive as \$12,000.00 are stripped to floats. Dates such as 31-Dec-04 are parsed with an explicit day-month-year format.

How to say this to a non-modeller. “Leverage” here is not debt over assets. It is “how much of this loan the government did not guarantee.” That is a real underwriting signal (weaker credits often have less guarantee) and it is *not* a balance-sheet ratio.

5.4 Attaching macros

GDP is quarterly. Unemployment is monthly in raw FRED and is resampled to quarter-start in the transform. Real GDP growth is a quarterly percent change, clipped to a plausible band so a level-vs-growth mix-up cannot explode the VAR.

Each loan receives the last available macro observation at or before its as-of date (`pandas.merge_asof`). Rows with a missing date are filled with the last FRED point and counted in the log (a few thousand on a full Kaggle extract).

LASSO is asked which lag of growth or unemployment helps forecast the next quarterly change in UNRATE. On this extract the cross-validated penalty sits at the top of the grid ($\alpha = 10$) and **every coefficient is zero**. The code still attaches `lag_1Q` so the PD feature list has names. That is a data fact, not a bug: origination-quarter unemployment does not line up with eventual SBA charge-off the way a textbook PIT model wants.

Chapter 6

Macro scenarios

6.1 What a scenario is here

A *row*: names of GDP growth and unemployment, plus the lag columns the PD model was trained on. Three rows:

	GDP growth (q/q)	Unemployment (%)	Construction
Baseline	VAR one-step (or last clean quarter)	same	statistical
Adverse	baseline plus configured shock	baseline plus shock	judgement
Severe	larger configured shock	larger shock	judgement

Default shocks in `config.yaml` are on the order of -2 pp / $+1.5$ pp (adverse) and -5 pp / $+3$ pp (severe) on growth and the unemployment *rate*. They are not EBA or Fed CCAR paths. They are knobs.

6.2 The VAR

Let $y_t = (\Delta \text{GDP}_t, U_t)'$. A $\text{VAR}(p)$ is

$$y_t = c + A_1 y_{t-1} + \dots + A_p y_{t-p} + \varepsilon_t. \quad (6.1)$$

p is chosen by AIC up to a small cap; if the information matrix is singular the fit falls back to $p = 1$ or to “repeat the last clean quarter.” The quarterly index is forced to a quarter-start frequency where possible so statsmodels does not warn about a missing calendar frequency.

A one-step forecast is the baseline. Shocks are *added* to that forecast. There is no 9-quarter path and no identification of structural shocks.

6.3 Why PD barely moves when you stress

The PD model is a cross-section HGB. On train, charged-off SBA loans do *not* sit at higher unemployment than paid-in-full loans (typical means $\approx 5.2\%$ vs 5.4%). So swapping baseline UNRATE for severe UNRATE changes very few leaves.

An earlier experiment added a logit overlay $\Delta \text{logit} = e_g \Delta \text{GDP} + e_u \Delta U$ with judgement elasticities, and another experiment estimated (e_g, e_u) from quarterly charge-off rates. The OLS signs on this tape were the wrong way ($e_u < 0$), so adverse PD *fell*. The overlay was removed. Scenario PD is the HGB output. Weak macro sensitivity is a **limitation**, not something the code hides.

Chapter 7

Probability of default

7.1 Label

$$Y_i = \mathbf{1}\{\text{MIS_Status contains CHGOFF}\} \vee \mathbf{1}\{\text{ChgOffPrinGr}_i > 0\}. \quad (7.1)$$

This is *ever charged off* on a resolved loan. It is not “defaulted in the next 12 months on a performing facility.”

7.2 Features

Default list: `leverage`, `liquidity_ratio`, `realgdp_growth_lag_1Q`, `unemp_lag_1Q`.

Monotonic constraints are mapped by *name* into HistGradientBoosting’s `monotonic_cst` vector: higher leverage $\Rightarrow$ higher PD; higher liquidity proxy $\Rightarrow$ lower PD. Macro signs are left unconstrained because the empirical relationship is weak.

7.3 Estimator

`sklearn.ensemble.HistGradientBoostingClassifier`, early stopping, `predict_proba[:,1]` as $\widehat{\text{PD}}_{12\text{m}}$.

A logistic regression challenger is fit on the same split. Holdout metrics (full Kaggle run, 30% test, stratified) were on the order of:

	n	Default rate	Mean $\hat{p}$	AUC
HGB	$\approx 269,000$	0.181	0.181	≈ 0.68
Logit	same	0.181	0.181	≈ 0.67

Calibration slope near 1. Deciles rise with predicted PD (low decile default rate $\approx 3\%$, top decile $\approx 30\%$). That is a real ranking on guarantee share and firm size. It is not a 0.85 AUC bureau model.

7.4 Why AUC is only moderate

The label is a lifetime resolution mixed across decades. Features are two proxies plus two macros the tree barely uses. There is no FICO, no current ratio, no delinquency history. 0.68 is consistent with that.

Chapter 8

Loss given default

8.1 Two-part model

LGD on a whole book is zero for most names (they do not default) and a fraction in $(0, 1]$ for names that do. The engine follows a standard hurdle:

1. Logit: $P(\text{loss severity} > 0 \mid x)$.
2. Ridge regression: $\mathbb{E}[\text{severity} \mid \text{severity} > 0, x]$.
3. $\widehat{\text{LGD}} = \hat{p}_{\text{hurdle}} \times \hat{s}_{\text{ridge}}$.

A downturn floor (config, default 15%) is applied in stress so LGD cannot collapse to the in-sample mean of a high-recovery book.

Observed SBA severity is `ChgOffPrinGr`/EAD on charge-offs. That is an accounting charge-off rate, not a discounted workout LGD with costs and timing.

Chapter 9

Staging and ECL mathematics

9.1 Scoring a scenario

For each scenario row s :

1. Copy the tape; overwrite the macro lag columns with the scenario values.
2. $PD_{i,s} = \text{HGB}(x_i^{\text{fin}}, x_s^{\text{macro}})$.
3. $LGD_{i,s} = \max(\text{downturn floor}, \text{hurdle} \times \text{ridge})$.
4. Stage i from SICR and DPD as in Chapter 3.
5. Term-structure ECL as below.

9.2 Discounted marginal ECL

Let $h_i = PD_{i,s}$ (annual hazard), r the discount rate (config 3%), M_i remaining maturity in years capped at $H = 5$, L_i LGD, E_i EAD. Survival starts at $S_{i,0} = 1$. For $t = 1, \dots, H$:

$$\text{live}_{i,t} = \mathbf{1}\{M_i \geq t - 1\}, \quad (9.1)$$

$$\phi_{i,t} = \min(1, \max(0, M_i - (t - 1))), \quad (9.2)$$

$$m_{i,t} = S_{i,t-1} h_i \phi_{i,t} \text{live}_{i,t}, \quad (9.3)$$

$$\text{ECL}_{i,t} = \frac{m_{i,t} L_i E_i}{(1 + r)^t}, \quad (9.4)$$

$$S_{i,t} = S_{i,t-1} (1 - h_i \phi_{i,t}). \quad (9.5)$$

$$\text{ECL}_i^{\text{life}} = \sum_{t=1}^H \text{ECL}_{i,t}, \quad \text{ECL}_i^{12\text{m}} = \text{ECL}_{i,1}. \quad (9.6)$$

Stage 1 uses the 12-month piece only. Stages 2 and 3 use the lifetime sum. Stage 3 can equivalently be thought of as $L_i E_i$ when PD is already near one; the code still runs the same recursion.

9.3 Probability-weighted ECL

$$\text{PWECL} = \sum_{s \in \{\text{base}, \text{adv}, \text{sev}\}} w_s \sum_i \text{ECL}_{i,s}, \quad \sum_s w_s = 1. \quad (9.7)$$

Coverage is $\sum_i \text{ECL}_{i,s} / \sum_i E_i$.

Chapter 10

What the engine actually printed

Figures move slightly across retrains. Orders of magnitude from the full Kaggle+FRED path:

Tape rows after clean	$\approx 897,000$
Default rate (CHGOFF)	$\approx 18.1\%$
Sum of EAD	$\approx \$181 \text{ bn}$
Median EAD	$\$100,000$
Mean maturity	$\approx 9 \text{ years (capped at 5 in ECL)}$
HGB holdout AUC	≈ 0.68

With **no PD overlay**, scenario PD stays near the HGB mean ($\approx 25\%$ on this resolved book). LGD can still move a little if GDP is in the LGD feature list. Stage mix is dominated by the SICR rule versus origination PD, not by a recession re-score.

How to read those dollars. A 4% coverage on a resolved historical tape is not a CET1-relevant bank allowance. It is “this formula applied to these proxies.” Compare runs to each other, not to a 10-Q.

Chapter 11

MLflow, models on disk, and GenAI

11.1 Two places models live

`models/pd_model.pkl, models/lgd_model.pkl`

What the engine loads. Joblib. Sufficient to rerun stress without MLflow.

`mlruns/`

Experiment tracking: parameters, holdout metrics, optional sklearn flavour with an input example. Inspect with `python3 -m mlflow ui --backend-store-uri ./mlruns --port 5000`.

Integer-column schema hints from MLflow are cosmetic.

11.2 What GenAI does—and does not

If `OPENAI_API_KEY` is set and you pass `--genai`:

- a LangChain pandas agent writes two sentences on the severe facility table;
- an optional RAG helper can quote SR 11-7-like text if a PDF is supplied.

Those strings are appended under “Notes.” They are not a control, not a PD, and not an audit opinion. Quota errors skip the block; ECL still runs. `Chain.run` is deprecated; the agent uses `invoke`.

Chapter 12

How to run it

```
cd macro-ecl
python3 -m venv venv
source venv/bin/activate          # Windows: venv\Scripts\activate
python3 -m pip install -r requirements.txt
# place SBAnational-scale CSV at data/raw/sba_raw.csv
# place FRED extract at data/raw/historical_macro_data.csv
PYTHONPATH=. python3 -m pytest -q
PYTHONPATH=. python3 scripts/run_all.py
```

If pip or mlflow complain about a bad interpreter, use `python3 -m pip` and `python3 -m mlflow`. Recreate the venv rather than repairing shebang paths after moving the folder.

Chapter 13

Limitations (leave these in)

1. **Resolved book.** Labels are PIF vs CHGOFF, not 12-month default on performing names. Mean PD $\approx 25\%$ is not a live SME PD.
2. **Proxy financials.** Leverage is the unguaranteed share. Liquidity is a transform of headcount. There are no GAAP statements.
3. **Macros start in 2000.** Originations from the 1970s–1990s have no proper origination macro. Pre-2000 SICR uses baseline PD as the anchor.
4. **LASSO selects nothing.** $\alpha = 10$, all coefficients zero. Lag_1Q is a fallback name.
5. **HGB does not reprice with UNRATE.** Stress PD barely moves. That is why the overlay was deleted instead of forced.
6. **Constant hazard lifetime.** Equation (3.2), cap five years. No prepayment, no competing risk, no time-varying hazard.
7. **Point scenarios.** One row each; not a multi-year path.
8. **Constant EAD, no CCF.** Undrawn lines do not exist on this file.
9. **Random holdout, not out-of-time.** AUC is in-sample in calendar time.
10. **DPD ≈ 0 .** Stage 2/3 are PD-rule driven.
11. **GenAI is commentary.**

A bank tape with performing exposures, financial statements, stored origination PD, and a monthly arrears history would use the *same equations* and produce a different economic picture. The code is written so those columns can be mapped without rewriting the engine.

Chapter 14

What “better data” would change—and what it would not

Public alternatives (Lending Club consumer loans; Freddie Mac monthly mortgage performance) would improve *some* of the list above (Current vs Charged Off; true LTV and monthly default). They still would not be a bank C&I book. This project stays on Kaggle SBA National by choice: it is the SME-shaped public file, and the limitations are documented rather than synthesised away.

Chapter 15

Glossary

As-of date

The date used to look up the last known macro observation.

CHGOFF

SBA charged-off.

EAD

Exposure at default.

ECL

Expected credit loss (IFRS 9 measurement).

FRED

Federal Reserve Economic Data.

HGB

Histogram-based gradient boosting (sklearn).

IFRS 9

IASB expected-loss accounting standard.

LGD

Loss given default.

PD Probability of default.

PIF

Paid in full.

PIT

Point-in-time (varies with the cycle).

PWECL

Probability-weighted ECL across scenarios.

SBA

U.S. Small Business Administration.

SICR

Significant increase in credit risk.

SR 11-7

Federal Reserve guidance on model risk management.

TTC

Through-the-cycle (averages over the cycle).

VAR

Vector autoregression.

Appendix A

Repository map

```
config/config.yaml
data/raw/sba_raw.csv
data/raw/historical_macro_data.csv
data/processed/sba_portfolio_tape.csv
data/scenarios/macro_forecasts.csv
models/pd_model.pkl
models/lgd_model.pkl
src/data_prep/clean_loan_tape.py
src/data_prep/sba_credit_pipeline.py
src/data_prep/transform_macro_data.py
src/macro_engine/var_model.py
src/macro_engine/macro_var_engine.py
src/ecl_models/train_pd_model.py
src/ecl_models/train_lgd_model.py
src/ecl_models/train_ecl_model.py
src/ecl_models/ecl_execution_engine.py
src/evaluation/metrics.py
src/evaluation/diagnostics.py
src/genAI_validation/sr117_report_generator.py
src/orchestrate.py
scripts/run_all.py
tests/
reports/ecl_macro_SR117_Final.pdf
```

Appendix B

Formula sheet

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD} \quad (\text{B.1})$$

$$\text{leverage}_i = 1 - \frac{\text{SBA_Appv}_i}{\text{GrAppv}_i} \quad (\text{B.2})$$

$$\text{PD}_{\text{life}}(T) = 1 - (1 - \text{PD}_{12\text{m}})^T \quad (\text{B.3})$$

$$\text{SICR} = \{\text{PD}/\text{PD}_0 \geq 2.5\} \vee \{\text{PD} - \text{PD}_0 \geq 0.08\} \vee \{\text{DPD} \geq 30\} \quad (\text{B.4})$$

$$\text{PWECL} = \sum_s w_s \text{ECL}_s \quad (\text{B.5})$$